ProjectHUD

+ CadenceHUD Workflow Engine

Architecture, Strategy & Integration Bible

Prepared by: Claude (Anthropic) — Systems Architecture Assessment
Date: April 1, 2026 | Version: 1.0 | Classification: CONFIDENTIAL

1. Executive Summary

ProjectHUD / CadenceHUD is a **unified operational intelligence platform for engineering-adjacent professional services firms.** It combines, in a single coherent application, capabilities that competing firms must assemble from 5–8 disconnected tools: project planning, personal work management, time & billing, meeting intelligence, process workflow control, and controlled document lifecycle management.

As of April 2026, the platform has reached **architectural completeness without operational proof.** The engines are built. The vision is demonstrable. The critical next phase is **closing the integration loops** — making the islands talk to each other in a way that delivers compounding value to real users on Day 1.

VERDICT	This platform, when integration loops are closed, will have no direct competitor in the professional services / engineering PM space. The moat is the unified audit trail + rework analytics + controlled document lifecycle — no one else has this combination.

2. Platform Architecture

The platform consists of three primary layers, each with distinct responsibilities but designed to share a single data model and audit trail.

2.1 Compass — The Daily Operations Layer

The user-facing personal productivity hub. Entry point for every team member, every day.

My Work	AI-prioritized daily work queue, KPI tiles, scored task recommendations, real-time time logging
My Time	Full timesheet management with PM approval workflow, billable/non-billable tracking, auto-narrative

My Calendar	Forward-only Kanban/Grid view — no rear-view mirror. Unfinished work always carries forward automatically
My Meetings	Living meeting minutes with persistent action items, threaded discussion, knowledge tree
My Views	Composable personal dashboards — infinite widget combinations, shareable, per-person or per-project
My Notes	Hierarchical organizational knowledge base, project-scoped, searchable
My Requests	CadenceHUD workflow catalog — browse, submit, and track any structured process request

2.2 ProjectHUD — The PM Execution Layer

MS Project-class project management with full EVM/CPM/PERT, multi-project dashboard, labor tracking, and critical path analysis.

Dashboard	Firm-wide KPIs: active projects, at-risk tasks, labor budget, EVM summary, team workload
Project Tasks	CPM/PERT with critical path, predecessor/successor, task status, duration, effort, % done
EVM / PERT	Earned Value Management with SPI/CPI, schedule variance, forecast completion
Gantt	Timeline visualization with dependency display
Labor	Budget vs. actuals, billability tracking per project and per resource
Risks / Docs	Risk register, document attachments, project journal

2.3 CadenceHUD — The Process Control Layer

Workflow micro-flow engine for any structured process that needs defined steps, role-based routing, approval chains, and an immutable audit trail.

Template Engine	Step/Diagram editor for defining workflow templates. Serial and parallel routing. Decision outcomes.
Instance Execution	Live instances with real-time status, elapsed time, rework tracking per step
Replay & Analytics	Timeline replay slider, rework cost panel (total loops, time in rework, per-step breakdown)
Form Engine	Controlled document lifecycle: PDF field mapping, role assignment, approval chain, CoC
My Requests	End-user catalog surface for submitting and tracking CadenceHUD workflow instances

3. The Integration Circuit — The Untapped Value Multiplier

Each layer has standalone value. The **exponential value** only activates when all three layers share a single data model and audit trail. This is the circuit that must be closed.

VISION	A project plan task fires a CadenceHUD workflow. That workflow contains form steps with controlled document lifecycle. All three events — task activation, workflow progress, form approval — appear in a single immutable CoC. No competitor in the PM space has this. It is the product's defining moat.

3.1 The Three Critical Integration Contracts

Contract 1: ProjectHUD Task → CadenceHUD Workflow (not yet formally defined)

- A project task must be able to declare an associated CadenceHUD workflow template
- Task completion should optionally gate on workflow instance completion
- Workflow launch should optionally be automatic (on task activation) or manual (user-initiated)
- Workflow completion should optionally update task status and log to task CoC

Contract 2: CadenceHUD Workflow → My Requests (UI shell exists, wiring partial)

- My Requests must be backed by real, live CadenceHUD templates — not a UI facade
- Submission must create a real CadenceHUD instance with proper firm/user context
- Status tracking in My Requests must reflect live instance state, not a separate status field
- Completion/rejection must surface in the submitter's My Work history

Contract 3: Notification / Inbox Model (must be standardized across all workflow types)

- When a workflow step lands in someone's queue, it MUST appear in My Work action items
- Email notification is secondary — My Work is the primary notification surface
- This model must be identical whether the trigger is a form review, PTO approval, or timesheet sign-off
- All notifications share the same infrastructure — no per-module notification silos

4. Current Platform Status

Honest assessment of each module as of April 2026. Status: LIVE = fully operational, PARTIAL = functional with gaps, STUB = UI exists but not wired, PLANNED = designed but not built.

Module / Feature	Status	Critical Gap / Next Action
Compass — My Work	PARTIAL	AI recommendation engine — confirm it runs on real data, not seed logic
Compass — My Time	PARTIAL	PM approval workflow — must use CadenceHUD flow, not standalone implementation
Compass — My Calendar	LIVE	Forward-only carry-forward working. Confirm capacity numbers are live.
Compass — My Meetings	LIVE	Living minutes with persistent AIs confirmed working. Knowledge tree search TBD.
Compass — My Views	PARTIAL	Widget library needs admin extensibility path defined
Compass — My Notes	PARTIAL	Notes not yet linked to project/meeting/task data model

Compass — My Requests	STUB	UI catalog exists. Must wire to real CadenceHUD templates end-to-end.
ProjectHUD — Tasks/EVM	LIVE	Core PM engine operational. Integration to CadenceHUD not yet defined.
ProjectHUD — Dashboard	LIVE	Multi-project KPIs working on seed data. Clear and re-test on real data.
CadenceHUD — Templates	PARTIAL	One workflow exists. Need 3+ templates to prove the model.
CadenceHUD — Instances	PARTIAL	Replay/CoC working. External user access (non-admin) not yet built.
CadenceHUD — Form Engine	PARTIAL	Definition layer polished. Runtime fill, released locking, evidence PDF missing.
Form — Reviewer Token Flow	STUB	Approval chain dialog built. External reviewer page (form-review.html) is a stub.
Form — Evidence PDF	PLANNED	Not yet built. Critical for regulated industry use cases.
Form — Released Locking	PLANNED	Must be enforced at save layer, not just UI button visibility.

5. Seed Data Clear & Go-Live Plan

The platform is currently running on **characterized seed data in Supabase** to facilitate design and development. Before first real-client engagement, this must be cleared and replaced with verified live data flows. This is the correct and necessary next step.

WARNING	Do NOT clear seed data until at least one complete vertical slice (see Section 6) is proven end-to-end with real users. Clearing too early removes the reference state you need to validate integration behavior.

5.1 Clear Sequence

1. Back up current Supabase state completely — snapshot all tables before any truncation
2. Prove Vertical Slice 1 (Document Review) end-to-end with at least 2 real users
3. Prove Vertical Slice 2 (PTO Request) end-to-end
4. Prove Vertical Slice 3 (Timesheet Approval) end-to-end
5. Clear seed data from all transactional tables (preserve schema, categories, and template definitions)
6. Re-run all three slices on clean data to confirm no hidden seed dependencies
7. Remove or gate the ClearHistory (⌫) dev button behind CADENCE_DEV flag before any client access

6. The Three Vertical Slices — Priority Build Order

A **complete vertical slice** means: a real user with no guidance can initiate something → it routes correctly → others get notified and can action their part → it completes → the result is visible in the right places → the audit trail is complete. Zero gaps.

Slice 1: Document Review & Sign-Off — PRIORITY 1

This slice proves the entire platform integration story and is the most commercially compelling demo for regulated-industry clients.

FLOW	My Requests → "Document Review & Sign-off" → CadenceHUD workflow fires → Form engine lifecycle runs → Reviewer gets token-authenticated link → Reviewer approves or rejects → Form releases → Evidence PDF generated → CoC visible in submitter's My Work history

Missing pieces to close this slice:

- External reviewer token flow (form-review.html stub must become real)
- Released form hard-locking at _formSave() layer (not just UI button visibility)
- Evidence PDF generation on form completion (flattened PDF + CoC audit block)
- My Requests wired to real CadenceHUD template (not facade)
- Completion event surfaces in submitter's My Work history

Slice 2: PTO / Leave Request — PRIORITY 2

This slice proves the everyday-user story — the unglamorous operational stuff that builds daily trust.

FLOW	My Requests → "PTO / Leave request" → Workflow routes to PM in My Work action items → PM approves or returns → Calendar blocks automatically in My Calendar → Team capacity updates in My Views

Missing pieces to close this slice:

- My Requests → real CadenceHUD template wiring
- PM approval action appearing in My Work (not separate inbox)
- Calendar auto-block on approval (My Calendar integration)
- Capacity recalculation in My Views on approval

Slice 3: Timesheet Approval — PRIORITY 3

This slice proves financial/billing trust and makes the platform operationally required, not optional.

FLOW	My Time → submit week → PM sees approval request in My Work action items → PM approves or returns with comments → Status updates in My Time → Billing records locked

Missing pieces to close this slice:

- Timesheet approval must use CadenceHUD flow (not standalone implementation) — prevents two parallel approval chains
- PM approval action appears in My Work (consistent with all other approvals)
- Approved timesheet records become locked / read-only

7. Architecture Decisions — Do Not Revisit

These design decisions are correct. They should not be simplified or second-guessed under feature pressure.

State machine (form lifecycle)	draft → in_review → reviewed → approved → released → unreleased with rejected_* substates. DB-constrained. Do not simplify. Regulated clients need this granularity.
CoC write pattern	Write CoC event BEFORE _formSave() with 50ms delay. This ordering is load-bearing. Changing it causes out-of-sequence activity table display.
Version pinning	Versions stay at 0.1.0 through all pre-release cycles. Only bump on Create Revision. Semantically correct for document control. Non-negotiable.
Forward-only calendar	My Calendar has no rear-view mirror. Unfinished work always carries forward. Strong, opinionated design choice that reduces cognitive load.
Field types (form engine)	review, doc_ref, attendees beyond the basics are domain-specific types that differentiate from generic form tools. Keep them.
Replay-capable CoC	The timeline replay slider with rework cost analytics is a genuine differentiator. No standalone workflow tool can compute rework cost — they don't know about the hours. Protect this capability.
My Work as primary inbox	All workflow notifications — form reviews, approvals, timesheets — must surface in My Work. Email is secondary. Never build per-module notification silos.

8. Competitive Positioning & Target Market

8.1 What No Competitor Has

This is the combination that creates the moat. Any single element exists elsewhere. The combination does not.

- Individual daily work management (tasks, time, calendar, capacity)
- Team/project execution management (CPM, EVM, Gantt, labor)
- Structured process control (workflows, approvals, CoC)
- Controlled document lifecycle (form definitions, versioning, review chains)
- Organizational knowledge management (meetings, decisions, notes)
- All of the above sharing a single, immutable, replay-capable audit trail

8.2 Target Market

Primary: Engineering-adjacent professional services firms who manage projects, people, and controlled processes simultaneously.

- Medical device consultancies (FDA/ISO compliance, IOPQ, design reviews)
- Aerospace engineering services (AS9100, design review, ITAR documentation)
- Contract R&D organizations (multi-client, billable hours, controlled deliverables)
- Quality management consultancies (QMS, CAPA, NCR workflows)

Displacement story: These firms currently spend $200–400/user/month across: MS Project / Smartsheet + Harvest/Toggl + DocuSign/Adobe Sign + Confluence + a QMS tool. ProjectHUD replaces all of them with something more capable because the data is unified.

8.3 Go-To-Market Wedge

Enter on **My Requests + the Document Review workflow.** Every professional services firm has someone whose job is to chase down approvals and document sign-offs. Show them that one workflow running flawlessly with full audit trail — then let them pull in the rest of the platform themselves.

WEDGE	"We can replace your document approval chase process with a controlled, auditable, replay-capable workflow — and it lives inside your project management tool." That sentence opens the door. The rest of the platform closes the deal.

9. Next-Generation Feature Roadmap

Prioritized by commercial impact and competitive differentiation. Do not build these until the three vertical slices are closed.

Tier 1 — Close Current Gaps (Table Stakes for Enterprise)

- External reviewer token flow — without this, approval chains are single-actor
- Released form hard-locking at save layer
- Evidence PDF generation — flattened, signed, CoC-stamped artifact on form completion
- In-app notification badges for pending reviews in My Work
- My Requests → CadenceHUD live wiring for all 9 request types

Tier 2 — Differentiation (6–12 months)

- Morning Brief AI synthesis — daily AI summary across all data streams: what's overdue, who's at risk, what needs a decision. When the underlying data is trusted, this becomes the platform's signature feature.
- Rework analytics dashboard — firm-level: which process types generate most rework, which reviewers are bottlenecks, cost of rework over time. This is a management-level sales story.
- Conditional form logic — fields that appear or become required based on other field values. Unlocks QMS use cases (NCR, CAPA workflows).
- Cross-form dependency chains — Form B cannot be submitted until Form A is Released. Models real engineering dependencies.
- Form template library — pre-built templates for AS9100, ISO 9001, FDA 21 CFR Part 11, OSHA. Zero-to-value in minutes.

Tier 3 — Strategic Moat (12+ months)

- Full task-flow-document CoC unification — when a project task triggers a flow that produces a controlled document, all three events appear in one audit trail.
- Mobile-native fill experience — iOS/Android for field technicians with offline support, camera-to-signature, push notifications.

- E-signature legal compliance layer — timestamped, IP-logged signatures integrated with a compliant e-sign provider. Removes major procurement objection in regulated industries.
- Client portal — external client view of project status, pending approvals, and deliverable sign-offs. Currently shown in nav but not detailed.

10. Known Issues — Form Engine

Carried forward from handoff documentation. All known and tracked.

_formRefreshUI() self-call	When _formPreviewMode is false, function calls itself recursively. Latent infinite-loop risk. Add explicit renderFormsTab(el) guard.
Dev clear-history button	⌫ button on left rail is exposed in production. Gate behind window.CADENCE_DEV === true before any client sees the app.
remaining confirm() calls	_formCreateRevision, _formArchiveForm, _formDeleteWithConfirm still use browser confirm(). Replace with modals matching Cancel Revision pattern.
Section header backgrounds	LIFECYCLE/ACTIVITY/COMMENTS headers need distinct background color (pending work item).
Button raised relief	Standardize box-shadow: 0 2px 4px rgba(0,0,0,.4), inset 0 1px 0 rgba(255,255,255,.08) across all action buttons.

11. Session Handoff Protocol

This document is the strategic bible. For each development session, a separate session handoff (the format used in the current project) carries the tactical state: version stamps, Supabase config, iron rules, pending changes, and bug patterns.

Two-document system:

- [object Object] — strategic architecture, never changes without deliberate review. Share with stakeholders, investors, clients.
- [object Object] — tactical state, updated every session. Paste into Claude at the start of each work session.

RULE	When a new insight, decision reversal, or architectural change is made in a development session, update this document before the session ends. A strategic document that is not maintained is worse than no document — it creates false confidence.

12. Closing Assessment

You have built the hardest parts. The **engines are real**, the architecture is sound, the vision is differentiated, and the islands are demonstrable. The gap is not capability — it is the **last-mile operationalization** that turns a sophisticated platform into something a normal person can use on Day 1 without a guide.

The sequence: **close the form reviewer loop → wire My Requests → build the first ProjectHUD-to-CadenceHUD task link → ship Evidence PDF.** That sequence, done in order, gives you a complete demo-able story and your first real client use case.

FINAL WORD	This platform, when the integration loops are closed, will have no direct competitor for engineering-adjacent professional services firms. The combination of replay-capable CoC + rework analytics + controlled document lifecycle inside a full PM platform is genuinely novel. Ship the three slices. Then the moat builds itself.